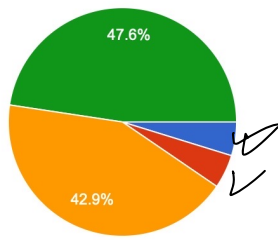


Preferred 1st date

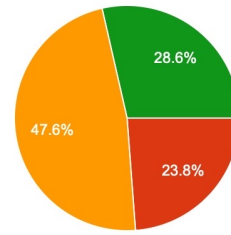
21 responses



3 June
4 June
5 June
6 June

Preferred 2nd date

21 responses



3 June
4 June
5 June
6 June

~ 6 6th June
~ 6 7th June
~ 2 6th Jun
~ 2 3rd June

1/7 → 25/7

June session 3/6 → 10/7

July session 11/7 → 25/7

26/5

27/5 8:30

28/5

CP violation and origin of mass in the Standard

Symmetry	C	P	CxP
QCD	✓	✓	✓
QED	✓	✓	✓
Weak	✗	✗	?

Goldhaber.

$$\nu \xrightarrow{\text{C}} \bar{\nu} \xrightarrow{\text{P}} \nu$$

C

$$\nu \xrightarrow{\text{C}} \bar{\nu} \xrightarrow{\text{P}} \nu$$

$$\nu \xrightarrow{\text{C}} \bar{\nu} \xrightarrow{\text{P}} \nu$$

CP is conserved in Goldhaber experiment

CP $\equiv$ matter $\leftrightarrow$ anti-matter transformation

Experimentally $A : a + b \rightarrow c + d$

$\bar{A} : \bar{a} + \bar{b} \rightarrow \bar{c} + \bar{d}$

1) $|A| = |\bar{A}|$?

2) distribution of c, d same as $\bar{c}, \bar{d}$?

T : time inversion

$$a + b \rightarrow c + d$$

$$c + d \rightarrow a + b$$

$$e^+ + e^- \rightarrow \gamma \gamma$$

$$\gamma + \gamma \rightarrow e^+ + e^-$$

CP T is a good symmetry (CPT theorem);

WU experiment
 $\gamma^+ (1 - \gamma^+)$

$$Co \rightarrow Ni + e^- + \bar{\nu}_e$$

Right and left are different



polarization of Co affects hemisphere of e^- emission.

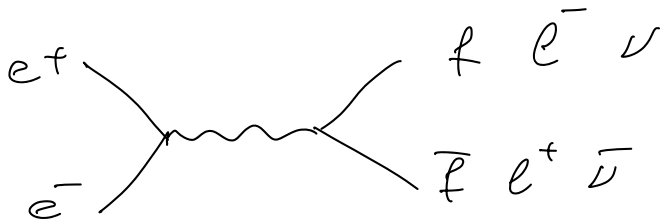
$C\bar{P} \rightarrow$ evidence for $\bar{X}$

$$\frac{N_{\bar{\text{baryons}}}}{N_{\text{baryons}}} = 10^{-6} - 10^{-6}$$

matter-anti-matter
asym. in the
universe

Experimentally measured? in space: AMS, Fermi;
on Earth (cosmic rays).

count $\# p, \# \bar{p}, \# He, \# \bar{He}, \dots$



lepton number conservation.

baryon number conservation.

why large asym in today's universe if $N_{\bar{B}} = N_B$
after Big Bang?

$$\frac{N_{\text{baryons}}}{N_{\text{photons}}} \approx 10^{-10}$$

1967 Sakharov how to explain the observed asym

- Baryon $\#$ violation
- non-equilibrium during evolution
- C and CP violation in weak, QCD, QED?

First evidence for $C\bar{P}$ 1964. (Cronin-Fitch exp.)

CP? $\bar{l}\bar{f} \quad C = (-1)^{L+S} = (-1)^{L+S}$

$$P = (-1)^{L+S} = (-1)^{L+S}$$

$$= (-1)^L \times (-1)(+1)$$

P of $\bar{l}\bar{f}$ in QFT

CP eigenstates? neutral electric charge.

$$\pi^+ \pi^-$$

$$\pi^+ \pi^- \pi^0$$

$$\pi^0 \pi^0$$

$$K^+ K^-$$

$$K^0 \bar{K}^0$$

(potential eigenstates)

CP : Start with $CP_i = +1 \xrightarrow{\text{Weak}} CP_f = -1$.

$$\langle f | H_{\text{weak}} | i \rangle \neq 0$$

$\Rightarrow CP$ in weak interaction.

Types of CP violation.

1) Direct CP violation.

$$A: i \rightarrow f$$

$$\bar{A}: \bar{i} \rightarrow \bar{f}$$

$$\#(i \rightarrow f) \neq \#(\bar{i} \rightarrow \bar{f})$$

$$|A(i \rightarrow f)| \neq |\bar{A}(\bar{i} \rightarrow \bar{f})|$$

Exp: Start with f^0, f^+
 $\bar{f}^0, f^-$

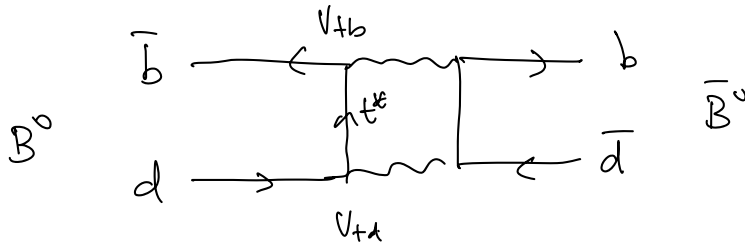
B^0, B^+

$\bar{B}^0, B^-$

2) CP violation in mixing

$$B^0 \leftrightarrow \bar{B}^0, K^0 \leftrightarrow \bar{K}^0, D^0 \leftrightarrow \bar{D}^0$$

$$B^0 \equiv \bar{b}d$$



$$\begin{array}{lcl} \bar{b}d & B^0 & \leftrightarrow \bar{B}^0 \\ \bar{c}d & \bar{D}^0 & \leftrightarrow D^0 \\ \bar{s}d & K^0 & \leftrightarrow \bar{K}^0 \end{array}$$

$$A_{CP} = \frac{N(B^0 \rightarrow \bar{B}^0) - N(\bar{B}^0 \rightarrow B^0)}{N(\bar{B}^0 \rightarrow D^0) + N(B^0 \rightarrow \bar{D}^0)} \in [0, 1]$$

experimentally: $A_{CP} \approx 10^{-4}$ Compatible with $\emptyset$

3) Interference between mixing and decay

$$B^0 \xrightarrow{A_1} f = \pi^+ \pi^-, \pi^0 \pi^0, K^0 \pi/\pi, D^0 \bar{D}^0, D^+ D^-, \dots$$

CP eigenstates

$$A_1: B^0 \rightarrow f$$

$$A_2: B^0 \rightarrow \bar{B}^0 \rightarrow f$$

$$\mathcal{M}(i \rightarrow f) = A_1(B^0 \rightarrow f) + A_2(B^0 \rightarrow \bar{B}^0 \rightarrow f)$$

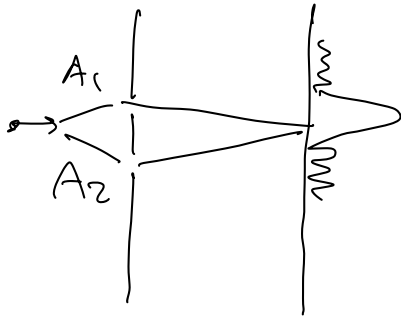
$$\#(i \rightarrow f) \propto |\mathcal{M}|^2 = |A_1|^2 + |A_2|^2 + A_1^* A_2 + A_2^* A_1$$

$$A_1 = a_1 e^{i\alpha_1} \quad A_2 = a_2 e^{i\alpha_2}$$

a_1, a_2 : real.

α_1, α_2 : Change under weak interaction.

$$\alpha_1 \neq \alpha_2$$



Weak interaction $q_i \rightarrow V_{ij} q_j$

$$V_{CKM} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}$$

CKM: complex numbers $\Rightarrow$ could cause interference between A_1, A_2

V_{CKM} must be unitary / hermitian.

$$V_{CKM}^\dagger V_{CKM} = \mathbb{1}$$

$$V_{CKM}^\dagger = \begin{pmatrix} V_{ud}^* & V_{cd}^* & V_{td}^* \\ V_{us}^* & V_{cs}^* & V_{ts}^* \\ V_{ub}^* & V_{cb}^* & V_{tb}^* \end{pmatrix}$$

$\Rightarrow$ 9 constraints on CKM elements.

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$$

$$V_{ud}^* V_{us} + V_{cd}^* V_{cs} + V_{td}^* V_{ts} = 0$$

$$\lambda \lambda \lambda^5 \times$$

$$V_{ub}^* V_{ud} + V_{cb}^* V_{cd} + V_{tb}^* V_{td} = 0$$

$$\lambda^3 \lambda^3 \lambda^3$$

$$V_{us}^* V_{ub} + V_{cs}^* V_{cb} + V_{ts}^* V_{tb} = 0$$

$$\lambda^4 \lambda^2 \lambda^2 \times$$

$$V_{ud}^* V_{td} + V_{us}^* V_{ts} + V_{ub}^* V_{tb} = 0$$

$$\lambda^3 \lambda^3 \lambda^3$$

$$V_{td}^* V_{cd} + V_{ts}^* V_{cs} + V_{tb}^* V_{cb} = 0$$

$$\lambda^4 \lambda^2 \lambda^2 \times$$

$$V_{ud}^* V_{cd} + V_{us}^* V_{cs} + V_{ub}^* V_{cb} = 0$$

$$\lambda \lambda \lambda^5 \times$$

accessible with $B^0/\bar{B}^0$ decays.

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$$

$$|V_{cd}|^2 + |V_{cs}|^2 + |V_{cb}|^2 = 1$$

$$|V_{td}|^2 + |V_{ts}|^2 + |V_{tb}|^2 = 1$$

$$1$$

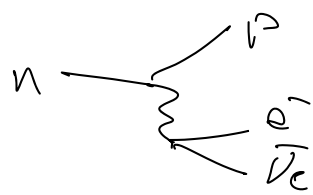
$$\emptyset$$

$$1$$

$$1$$

$$V_{CKM} = \begin{pmatrix} 1 - \lambda^2/2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \lambda^2/2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$

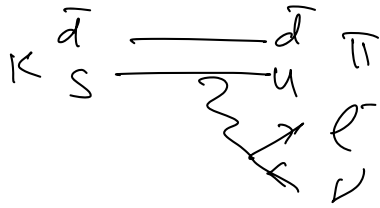
$$|V_{ud}|^2 \approx ?$$



β decay.

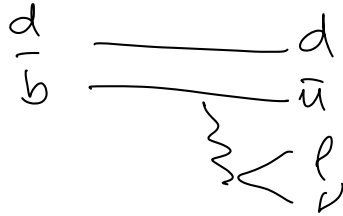
$$n \rightarrow p e^- \bar{\nu}_e$$

$$|V_{us}|^2 \approx ?$$



$$\bar{K}^0 \rightarrow \pi^+ e^- \bar{\nu}_e$$

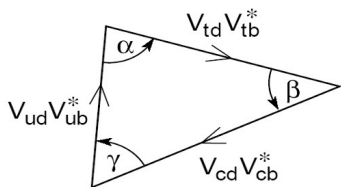
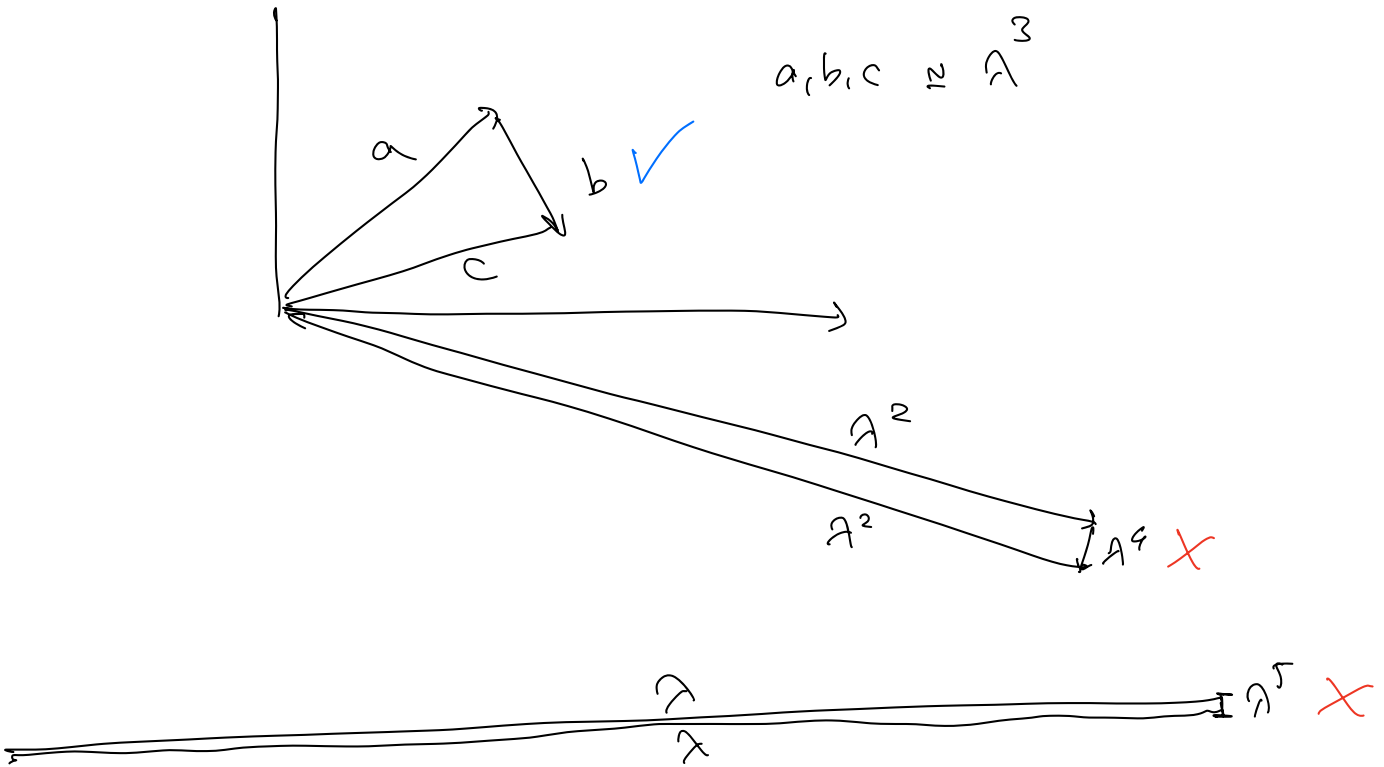
$$|V_{ub}|^2 \approx ?$$



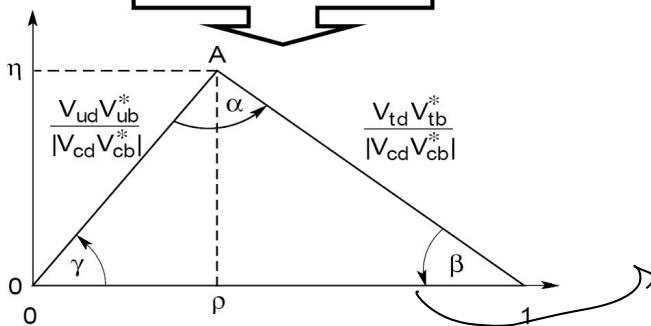
$$B^0 \rightarrow \pi^+ e^- \bar{\nu}_e$$

$$a + b + c = 0$$

$$a, b, c \approx \Lambda^3$$



Rescaling, aligning



$$V_{ud} V_{ub}^* + V_{td} V_{tb}^* + V_{cd} V_{cb}^* = 0$$

$$V_{CKM} = \begin{pmatrix} 1 & 1 & e^{-i\delta} \\ 1 & 1 & 1 \\ e^{-i\beta} & 1 & 1 \end{pmatrix}$$

$\hookrightarrow$

V_{td}

$$\frac{V_{cd} V_{cb}^*}{|V_{cd} V_{cb}^*|}$$

if you measure $\alpha \neq 0, \beta \neq 0, \gamma \neq 0 \Rightarrow$ triangle

$\Rightarrow$ CKM matrix has complex elements.

$\Rightarrow$ CP could be explained by CKM matrix
in weak
interactions.